

A hyperbolic approximation of the Cahn-Hilliard equation

Firas Dhaouadi

Università degli Studi di Trento

Joint work with

Sergey Gavriluk (Aix-Marseille University)

Michael Dumbser (University of Trento)



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3C: Challenges in Computational methods for Complex environmental applications

Cahn-Hilliard equations (1958)

The Cahn-Hilliard equation writes

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

- $c \in [-1, 1]$ is the order parameter indicating the phases.
- $\gamma \ll 1$ is such that $\sqrt{\gamma}$ is the diffuse interface characteristic length.
- describes phase separation in binary systems: spinodal decomposition, Ostwald Ripening phenomena, etc.

About the equation

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

Cool features

- scalar PDE.
- Well-posed.
- diffuse-interface model (able to deal with strong topological changes).

About the equation

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

Cool features

- scalar PDE.
- Well-posed.
- diffuse-interface model (able to deal with strong topological changes).

Not so cool features

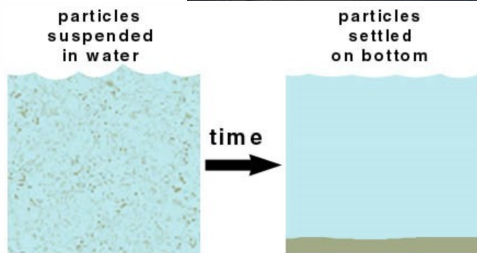
- non-convex energy potential (Requires very careful treatment)
- 4th Order in space (Forget about explicit solvers)
- Violates principle of Causality (Laplace operator)

Some environmental applications

1) Modeling of air pollution (Mixtures of gas/smoke)



2) Sedimentation problems



Plan of presentation

- 1 On the Cahn-Hilliard equations
- 2 Hyperbolic Model Derivation
 - 2nd-order approximation
 - 1st-order approximation approximation
 - Analysis
- 3 Numerical scheme and Results
 - Numerical schemes
 - Numerical results
 - Conclusion

Conservative form and chemical potential

The Cahn-Hilliard equation can be cast into a conservation-law form which writes

$$\frac{\partial c}{\partial t} + \operatorname{div}(\mathbf{j}) = 0, \quad (1)$$

where the mass flux $\mathbf{j}$ is assumed to obey a generalized Fick's law such that

$$\mathbf{j} = -\nabla \mu,$$

and μ is the chemical potential of the system given by

$$\mu = \frac{\delta f}{\delta c} = \frac{\partial f}{\partial c} - \operatorname{div} \left(\frac{\partial f}{\partial \nabla c} \right) = c^3 - c - \gamma \Delta c,$$

where

$$f(c, \nabla c) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2,$$

Lyapunov functional

CH equation admits the Lyapunov functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\Omega$$

Indeed, we have

$$\frac{\partial f}{\partial t} + \operatorname{div}(\mathcal{H}) = -\|\nabla \mu\|^2, \quad \mathcal{H} = -\mu \nabla \mu - \frac{\partial f}{\partial \nabla c} \Delta \mu,$$

which in integral form writes

$$\frac{\partial F}{\partial t} = - \int_{\mathcal{D}} \|\nabla \mu\|^2 \, d\Omega \leq 0.$$

Hyperbolic reformulation

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Modified action functional

Let us introduce the following action functional

$$a = \int_t \int_{\mathcal{D}} \mathcal{L} \, d\mathcal{D} dt$$

where

$$\mathcal{L} \left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t} \right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t} \right)^2.$$

- φ is a new variable substituting c as the order parameter.
- $\alpha \gg 1$ so that $(c - \varphi)$ vanishes in the limit $\alpha \rightarrow +\infty$.
- $\beta \ll 1$ is a small parameter.

Generalized Fick's law for c

$$\mathcal{L}\left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2}(c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

Generalized Fick's law now becomes

$$\frac{\partial c}{\partial t} + \operatorname{div}(-\nabla \mu) = 0, \quad \mu = -\frac{\delta \mathcal{L}}{\delta c} = -\frac{\partial \mathcal{L}}{\partial c} = c^3 - c + \alpha(c - \varphi),$$

$\Rightarrow$ 2nd-order PDE, no 4th-order terms

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

Euler-Lagrange equation for φ

$$\mathcal{L}\left(\mathbf{c}, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (\mathbf{c} - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

For φ , we simply write the Euler-Lagrange equations.

$$\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \varphi_t} \right) + \operatorname{div} \left(\frac{\partial \mathcal{L}}{\partial \nabla \varphi} \right) = \frac{\partial \mathcal{L}}{\partial \varphi}.$$

which gives

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div} (\gamma \nabla \varphi) = \alpha (c - \varphi) \quad (II)$$

2nd-order approximation of the Cahn-Hilliard equation

Thus so far we have obtained the following system of two 2nd order PDEs

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

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- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.

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- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.
- Equation (II) is a hyperbolic wave equation with right-hand side. $\Rightarrow$ Order reduction.

Order reduction for (II)

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div}(\gamma \nabla \varphi) = \alpha(c - \varphi) \quad (II)$$

Let us denote the independent variables

$$w = \beta \frac{\partial \varphi}{\partial t}, \quad \mathbf{p} = \nabla \varphi.$$

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Therefore (II) becomes

$$\begin{aligned} \frac{\partial w}{\partial t} - \operatorname{div}(\gamma \mathbf{p}) &= -\alpha(\varphi - c), \\ \frac{\partial \varphi}{\partial t} &= \frac{1}{\beta} w, \\ \frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w &= 0. \end{aligned}$$

Relaxation for equation (I)

$$\begin{aligned}\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) &= 0, \\ \frac{\partial \mathbf{q}}{\partial t} + \nabla \mu &= -\frac{1}{\tau} \mathbf{q},\end{aligned}$$

- $\tau \ll 1$ is a relaxation time.
- c is still a conserved quantity in this framework.

Final system approximating the Cahn-Hilliard equations

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla (c^3 - c + \alpha(c - \varphi)) = -\frac{1}{\tau} \mathbf{q}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} (\gamma \mathbf{p}) = -\alpha(\varphi - c)$$

$$\frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{1}{\beta} w$$

- System of hyperbolic equations with relaxations.
- Equations are conservative also in multiple dimensions.

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- System of hyperbolic equations with relaxations.
- Equations are conservative also in multiple dimensions.

(Has curl involutions on both $\mathbf{q}$ and $\mathbf{p}$ if you want to test curl-free schemes ...)

Hyperbolicity

System admits a full set of real eigenvalues ($\alpha > 1$) given by

$$\lambda_1 = -\frac{\sqrt{3c^2 + \alpha - 1}}{\sqrt{\tau}},$$

$$\lambda_2 = -\frac{\sqrt{\gamma}}{\sqrt{\beta}},$$

$$\lambda_{3-7} = 0,$$

$$\chi_8 = \frac{\sqrt{\gamma}}{\sqrt{\beta}},$$

$$\lambda_9 = \frac{\sqrt{3c^2 + \alpha - 1}}{\sqrt{\tau}}.$$

and a corresponding set of linearly independent eigenvectors. (easily computed explicitly).

Lyapunov Functional

Proposition

The proposed hyperbolic Cahn-Hilliard system admits the following Lyapunov functional

$$E = \int_{\mathcal{D}} e(c, \varphi, \mathbf{q}, \mathbf{p}, w) \, d\Omega,$$
$$e(c, \varphi, \mathbf{p}, w) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\mathbf{p}\|^2 + \frac{\alpha}{2} (c - \varphi)^2 + \frac{1}{2\beta} w^2 + \frac{1}{2\tau} \|\mathbf{q}\|^2$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi}$$

$$\frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w}$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial e}{\partial c} \cdot \left\{ \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

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$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

$$\Rightarrow \frac{\partial e}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial c} \frac{\partial e}{\partial \mathbf{q}} - \frac{\partial e}{\partial \mathbf{p}} \frac{\partial e}{\partial w} \right) = - \left\| \frac{\partial e}{\partial \mathbf{q}} \right\|^2 \leq 0,$$

Numerical methods

In order to solve the model numerically and also compare it with reference solutions, we propose here:

- ① A numerical scheme for the original Cahn-Hilliard equation based on 4th order semi-implicit conservative finite differences
- ② Explicit MUSCL-Hancock for the hyperbolic approximation.

Implicit conservative finite differences for CH

We propose here a semi-implicit conservative in order to solve numerically the original Cahn-Hilliard equations. We rewrite the latter as follows

$$\frac{\partial c}{\partial t} - \operatorname{div}(\mathbf{F}) + \gamma \Delta^2 c = 0$$

where $\mathbf{F}$ is the flux given by

$$\mathbf{F} = \chi(c) \nabla c, \quad \chi(c) = 3c^2 - 1$$

The scheme writes

$$c_{i,j}^{n+1} = c_{i,j}^n + \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j}^{n+1} - \mathcal{F}_{i-\frac{1}{2},j}^{n+1} \right) + \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}}^{n+1} - \mathcal{G}_{i,j-\frac{1}{2}}^{n+1} \right) - \gamma \Delta \Delta_h c_{i,j}^{n+1}.$$

Computation of the intercell fluxes

The intercell fluxes $\mathcal{F}_{i+\frac{1}{2},j}^{n+1}$ and $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$, in the x and y directions respectively, are computed using conservative finite-differences as follows

$$\mathcal{F}_{i+\frac{1}{2},j}^{n+1} = \chi_{i+\frac{1}{2},j}^n (\nabla_x c)_{i+\frac{1}{2},j}^{n+1},$$

$$\begin{cases} \chi_{i+\frac{1}{2},j}^n \simeq \frac{1}{12} (7\chi_{i,j}^n - \chi_{i-1,j}^n + 7\chi_{i+1,j}^n - \chi_{i+2,j}^n) \\ (\nabla_x c)_{i+\frac{1}{2},j}^{n+1} \simeq -\frac{1}{12\Delta x} (15c_{i,j}^{n+1} - 15c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} - c_{i-1,j}^{n+1}) \end{cases}$$

(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

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(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

These are 4th order approximations.

Discretization of the bi-Laplacian operator

$\Delta\Delta_h c_{i,j}^{n+1}$ is a discretization of the bi-Laplacian operator in the cell-centers as follows

$$\begin{aligned}\Delta\Delta_h c_{i,j}^{n+1} = & -\frac{1}{\Delta x^4} \left(c_{i-2,j}^{n+1} - 4c_{i-1,j}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} \right) \\ & -\frac{1}{\Delta y^4} \left(c_{i,j-2}^{n+1} - 4c_{i,j-1}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i,j+1}^{n+1} + c_{i,j+2}^{n+1} \right) \\ & -\frac{2}{\Delta x^2 \Delta y^2} \left(c_{i-1,j-1}^{n+1} - 2c_{i,j-1}^{n+1} + c_{i+1,j-1}^{n+1} - 2c_{i-1,j}^{n+1} \right. \\ & \quad \left. + 4c_{i,j}^{n+1} - 2c_{i+1,j}^{n+1} + c_{i-1,j+1}^{n+1} - 2c_{i,j+1}^{n+1} + c_{i+1,j+1}^{n+1} \right)\end{aligned}$$

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$$c_{i,j}^{n+1} = c_{i,j}^n + \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j}^{n+1} - \mathcal{F}_{i-\frac{1}{2},j}^{n+1} \right) + \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}}^{n+1} - \mathcal{G}_{i,j-\frac{1}{2}}^{n+1} \right) - \gamma \Delta\Delta_h c_{i,j}^{n+1}.$$

is then solved using GMRES (Matrix is not symmetric ...)

Numerical method for hyperbolic approximation

- Explicit second-order MUSCL-Hancock scheme
- We use either FORCE or Rusanov approximate Riemann solvers (One could also implement a Roe solver)

Exact solution for the original equation

One can find a family of exact one-dimensional stationary periodic solutions to the Cahn-Hilliard (Also Allen-Cahn) system expressed as

$$c_{\epsilon}(x) = \sqrt{1-\epsilon} \operatorname{sn} \left(\sqrt{\frac{\epsilon+1}{2\gamma}}(x-x_0), \sqrt{\frac{1-\epsilon}{1+\epsilon}} \right)$$

Here, $\operatorname{sn}(x, s)$ is the Jacobi elliptic sine function, and s is the elliptic modulus.
 $\epsilon \in [0, 1]$.

It is worthy of note that in the limit $\epsilon \rightarrow 0$ corresponding to $s \rightarrow 1$, one recovers the well-known solution

$$c(x) = \tanh \left(\frac{x-x_0}{\sqrt{2\gamma}} \right)$$

as a particular case.

Exact elliptic function solution

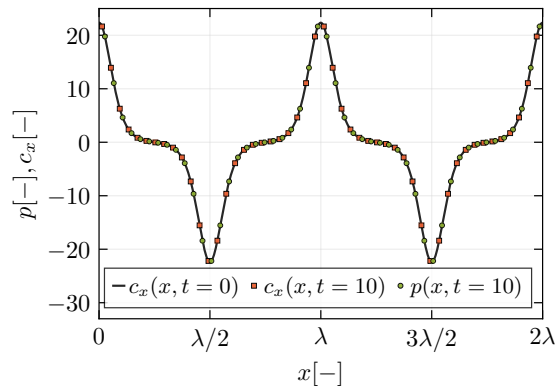
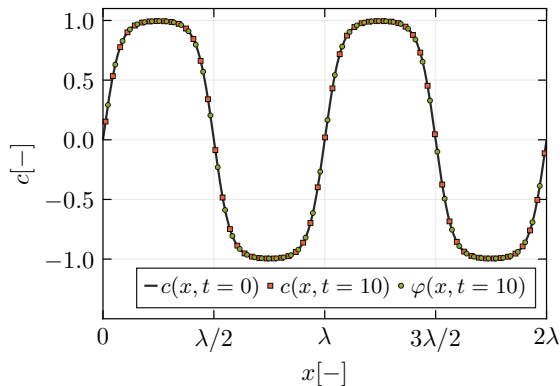


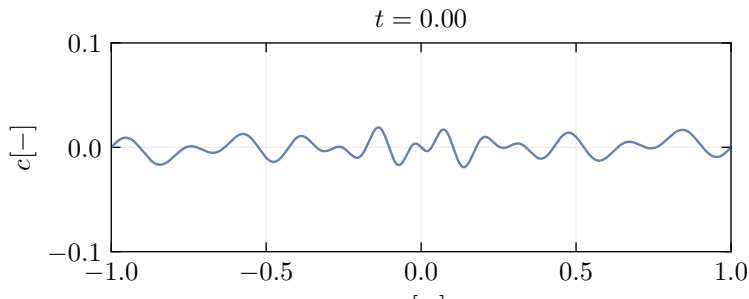
Figure 1: $\gamma = 0.001$. Computational domain is $[0, 2\lambda]$, discretized over $N = 2000$ cells. $\beta = 10^{-6}$, $\alpha = 500$ and $\tau = 8 \cdot 10^{-4}$. CFL = 0.95 and final simulation time is $t = 10$.

Spinodal decomposition

We suggest the following initial data

$$c(x) = \begin{cases} 0.01 \left((\sin(10\pi(1+x))) - \sin(10\pi(1+x)^2) \right), & \text{if } x \in [-1, 0] \\ -0.01 \left((\sin(10\pi(1-x))) - \sin(10\pi(1-x)^2) \right), & \text{if } x \in [0, 1]. \end{cases}$$

This function is built in such a way that it is C^∞ over $[-1, 1]$ as well as over $\mathbb{R}$ by periodic prolongation.



Spinodal decomposition ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)

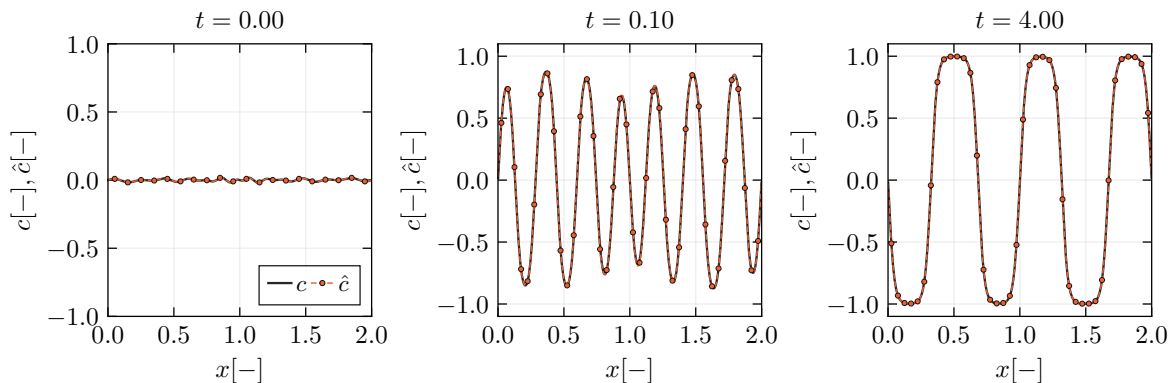


Figure 2: Comparison of the numerical results between the original model (orange) and its hyperbolic counterpart (black). $N = 2000$ computational cells.

Spinodal decomposition - Lyapunov function evolution ($\simeq 6b$ iterations)

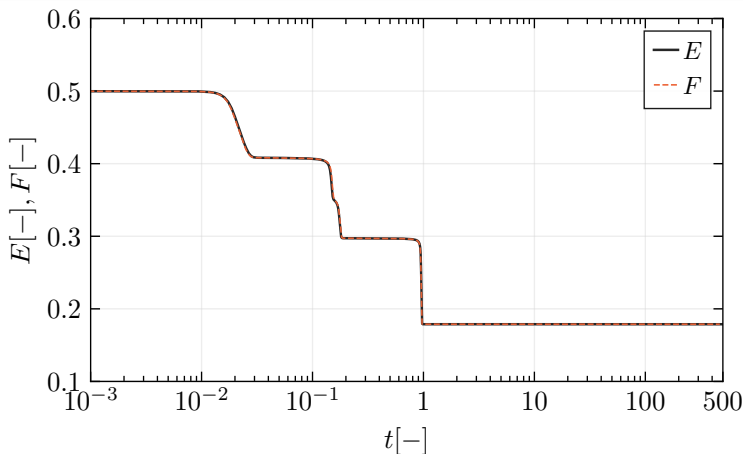


Figure 3: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Spinodal decomposition - Lyapunov function evolution ($\simeq 30b$ iterations)

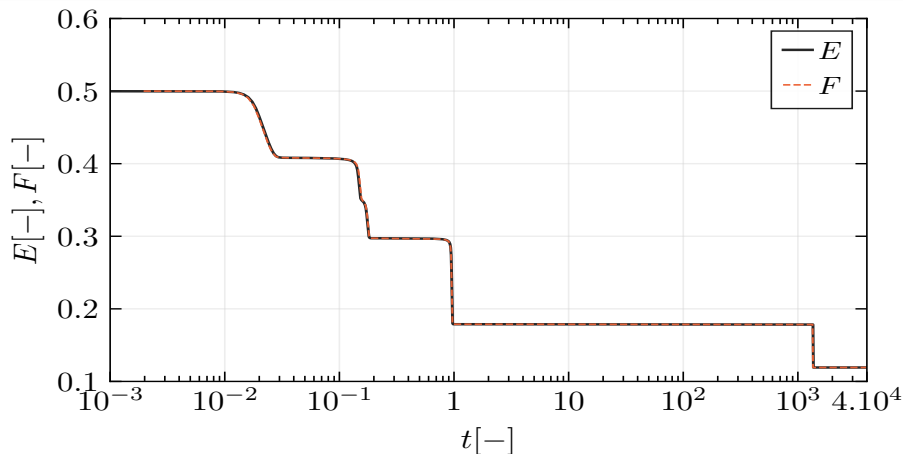


Figure 4: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Ostwald Ripening in 1D ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-4}$)

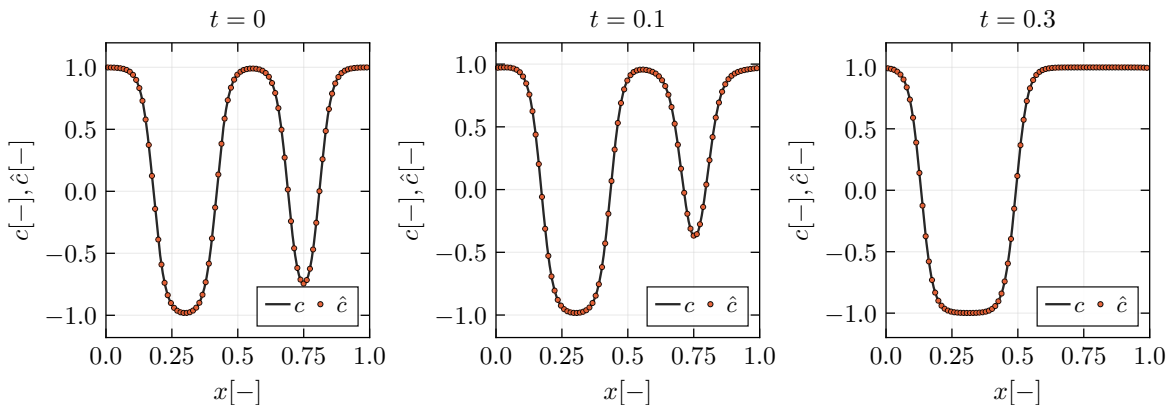
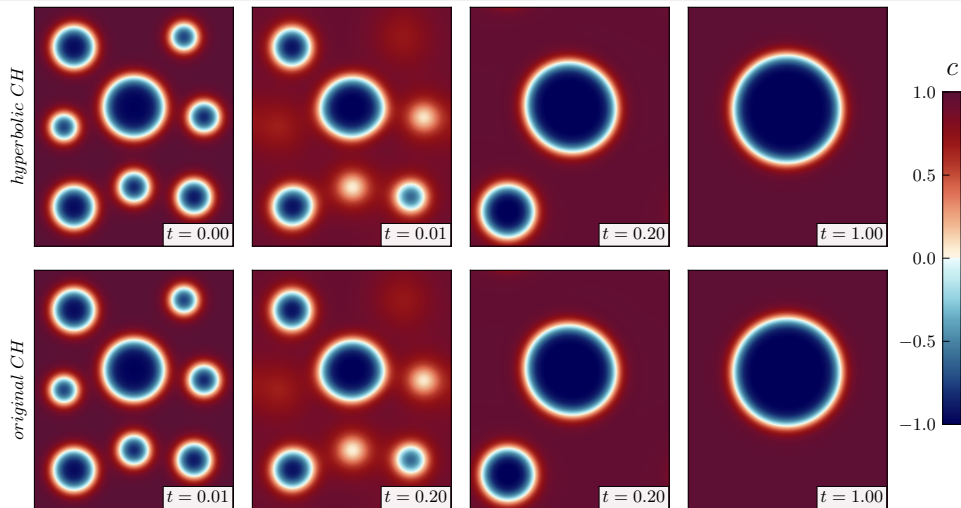


Figure 5: Comparison of the numerical solutions for hyperbolic Cahn-Hilliard model (black line) and the original model (red dots) for the Ostwald Ripening test case at times $t = \{0, 0.1, 0.3\}$.

Ostwald Ripening in 2D($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)



Ostwald Ripening in 2D : horizontal Cuts

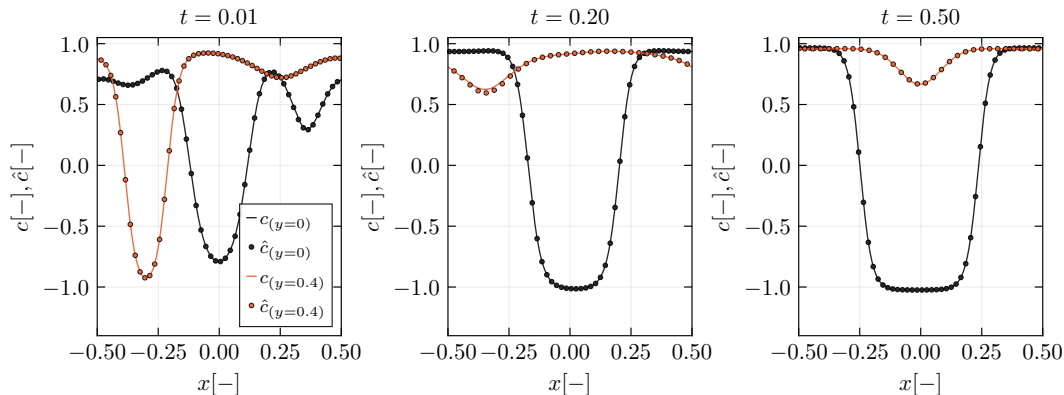


Figure 6: Horizontal cuts over the lines $y = 0$ (red) and $y = 0.4$ (black). Domain is 600×720

Conclusion and Perspective

- We presented a new formulation for an approximate hyperbolic Cahn-Hilliard system.
- An original scheme was conceived to solve the original equation using conservative finite differences.
- Comparison of results showed excellent agreement between the results in one and two dimensions.

Perspectives

- Better formulation fully from variational principles if possible.
- Extension to Navier-Stokes Cahn-Hilliard systems.
- Investigation of bound-preserving properties.
- Semi-implicit discretization, asymptotic preserving schemes, time-step optimization, etc ...

Thank you for your attention !

The article:

[1a] Dhaouadi F, Dumbser M, Gavriluk S. 2025 A first-order hyperbolic reformulation of the Cahn-Hilliard equation. **Proceedings of the Royal Society A**, 20240606. (10.1098/rspa.2024.0606)

The code:

[1b] Dhaouadi F, Dumbser M, Gavriluk S. 2025 Supplementary material from A first-order hyperbolic reformulation of the Cahn-Hilliard equation. The Royal Society. Collection. (10.6084/m9.figshare.c.7637355.v1)

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